2.4 Strong Maximum Principle

Notation: $\forall P_0 \in \bar{D} \setminus \Gamma$

$$S(P_0) \triangleq \left\{ Q \in \bar{D} \setminus \Gamma \;\middle|\; \exists \text{ continuous path } \gamma \text{ in } \bar{D} \setminus \Gamma, \text{ connecting } P_0 \text{ and } Q, \text{ and when traveling along } \gamma \text{ from } Q \text{ to } P_0, \text{ its } t\text{-coordinate is non-decreasing} \right\}.$$



$C(P_0) = S(P_0) \cap \{t = t_0\}$, $P_0 = (t_0, x_0)$

Rk Both $S(P_0)$ & $C(P_0)$ are connected because they are path-connected

SMP: $Lu = u_t - \left[a_{ij}(x,t)u_{x_i x_j} + b_i(x,t)u_{x_i} + c(x,t)u\right]$, $(x,t) \in D \subseteq \mathbb{R}^{n+1}$ (bdd domain)

Suppose L is strictly parabolic with a_{ij}, b_i, c bdd in D

Let $u \in C^{2,1}(D \setminus \Gamma) \cap C(\bar{D})$ satisfy $Lu \leq 0$ in $D \setminus \Gamma$

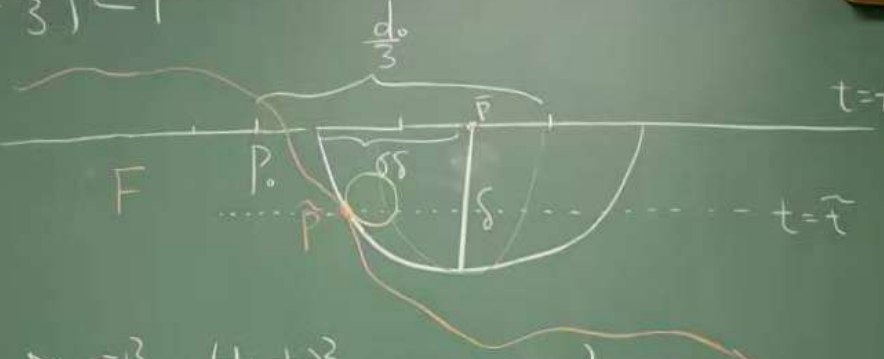
If $\exists\, P_0 \in D \setminus \Gamma$ s.t. $u(P_0) = \max_{\bar{D}} u$, then $u \equiv u(P_0)$ on $S(P_0)$,

provided (i) $c \equiv 0$ in D; or (ii) $c \leq 0$ in D, $u(P_0) \geq 0$; or (iii) $u(P_0) = 0$.

Pf Let $F = \{(x,t) \in \bar{D} \mid u(x,t) = u(P_0)\}$ W.T.S. $S(P_0) \subseteq F$

$d_0 \triangleq \text{dist}(P_0, \Gamma) > 0$ ($\because \Gamma$ is cpt, $P_0 \notin \Gamma$)

Claim 1 $B_{\frac{d_0}{3}}(P_0) \triangleq \{(x,t) \mid |x - x_0| < \frac{d_0}{3}\} \subset F$



Otherwise, $\exists\ \bar{P} = (\bar{x}, t_0) \in B_{\frac{d_0}{3}}(P_0)$

s.t. $u(\bar{P}) < M$

Let $\delta = \frac{1}{2}\text{dist}(\bar{P}, F)$

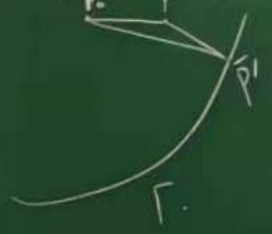
Semi-ellipsoid $E_\sigma \triangleq \left\{(x,t) \;\middle|\; \frac{|x - \bar{x}|^2}{(\sigma\delta)^2} + \frac{(t - t_0)^2}{\delta^2} \le 1,\ t \le t_0\right\}$, $\sigma > 0$.

If $0<\sigma<1$, then $E_\sigma \cap F = \phi$ $(\because \operatorname{dist}(\bar{P}, F) = 2\delta)$

If $\sigma\delta \geq \frac{d_0}{3}$, then $P_0 \in E_\sigma \Rightarrow E_\sigma \cap F \neq \phi$ $(\because P_0 \in F)$

Then increasing σ, we have that E_σ touches F before touching Γ.

Why:



$$\operatorname{dist}(\bar{P}, \Gamma) = |\bar{P}\bar{P}'| \geq |P_0\bar{P}'| - |P_0\bar{P}| \geq \operatorname{dist}(P_0, \Gamma) - |P_0\bar{P}|$$

$$\geq d_0 - \frac{d_0}{3} = \frac{2}{3}d_0 > \frac{1}{3}d_0 \geq |P_0\bar{P}| \geq \operatorname{dist}(\bar{P}, F)$$

Let the touching pt be $\hat{p} = (\hat{x}, \hat{t}) \in F$. We can put a ball B inscribed in E_σ, tangent to E_σ at $\hat{p}$. Then apply Hopf $\Rightarrow$

$\forall \vec{\eta}$ pointing outward to E_σ at $\hat{p}$, $\frac{\partial U}{\partial \vec{\eta}}(\hat{p}) > 0 \Rightarrow \nabla_x U(\hat{p}) \neq 0$

$\rightarrow\!\!\!\!\times$ $\hat{p}$ is an interior max pt in $\hat{t}$-level.

Claim 2 $C(p_0) \subset F$

Claim 1 $\Rightarrow F \cap C(p_0)$ is relatively open in $C(p_0)$

F closed $\Rightarrow F \cap C(p_0)$ is relatively closed in $C(p_0)$

$C(p_0)$ is connected, $F \cap C(p_0) \neq \phi$

$\Rightarrow F \cap C(p_0) = C(p_0)$

$\Rightarrow C(p_0) \subseteq F$

Claim 3. $U \equiv M$ on $S(P_0)$.

Otherwise, $\exists\, Q=(x_2,t_2) \in S(P_0)$ $(t_2<t_0)$ s.t. $U(Q)<U(P_0)$, i.e., $Q \notin F$.

Let $P_1=(x_1,t_1)$ be the first intersection of γ (a path connecting Q with P_0) with F when going upwards. Then on arc $\widehat{QP_1}$, $U<U(P_0)$

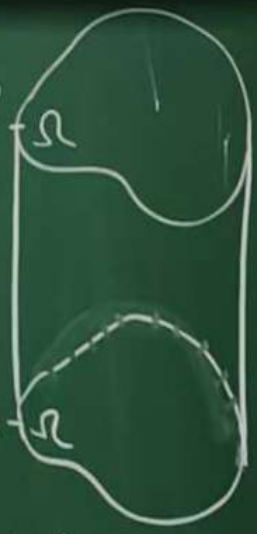
(Claim 2 $\Rightarrow U<U(P_0)$ on $S(P_0) \cap \{t_2 \le t < t_1\}$

$\Rightarrow$ The situation in previous Lemma happens $\longrightarrow\!\!\!\!\!\!\times$

□

Comparison Principle: $\Omega \subseteq \mathbb{R}^n$ bdd domain with $\partial\Omega \in C^2$.

$D = \Omega \times (0,T]$, $S = \partial\Omega \times (0,T]$, $\Gamma = S \cup \overline{\Omega}$



Let $Lu = u_t - a_{ij}(x,t)u_{x_ix_j} + b_i(x,t)u_{x_i} + f(x,t,u)$, $(x,t) \in \overline{D} \setminus \Gamma$,

$Bu = \frac{\partial u}{\partial \vec{\jmath}}(x,t) + \beta(x,t)u$, $(x,t) \in S$, $\beta \geq 0$ on S,

$\vec{\jmath}$ any outward pointing vector field on S

- L is strictly parabolic, & a_{ij}, b_i bdd in D,
- $f_s(x,t,s)$ exists for $s \in (-\infty, \infty)$, $(x,t) \in \overline{D} \setminus \Gamma$, and

$\forall R > 0$, $\exists$ const. $M = M(R)$ s.t. $|f_s(x,t,s)| \leq M$, $\forall (x,t) \in \overline{D} \setminus \Gamma$, $|s| \leq R$

Assume $u, v \in C^{2,1}(\bar{D})$ s.t.

$$\begin{cases} Lu \geq Lv \text{ in } \bar{D} \setminus \Gamma, \\ Bu \geq Bv \text{ on } S, \\ u|_{t=0} \geq v|_{t=0} \text{ on } \Omega. \end{cases} \Longrightarrow u \geq v \text{ in } \bar{D}.$$

Moreover, if $u|_{t=0} \not\equiv v|_{t=0}$, then $u > v$ in $\bar{\Omega}$ if $t > 0$.

Rk 1 : $\beta \equiv 0$ on S is allowed.

Rk 2 This also holds if $Bu = u$ on S ("$x \in \bar{\Omega}$" replaced by "$x \in \Omega$").

Pf: Let $w = v - u$. W.T.S. $w \leq 0$ in $\bar{D}$.

$$w_t - a_{ij}(x,t) w_{x_i x_j} + b_i(x,t) w_{x_i} + \underbrace{f(x,t,v) - f(x,t,u)}_{= f_s(x,t,\xi)\,(v-u)} = Lv - Lu \leq 0 \text{ in } \bar{D} \setminus \Gamma.$$

$$= c(x,t) \cdot w, \qquad |c(x,t)| \leq M$$

$\xi(x,t)$ between $v(x,t)$ & $u(x,t)$.

Let $\tilde{w} = e^{-Mt} \cdot w$.

$$\Rightarrow \tilde{w}_t - \left\{ a_{ij} \tilde{w}_{x_i x_j} - b_i \tilde{w}_{x_i} \underbrace{- (c(x,t) + M)}_{\leq 0} \tilde{w} \right\} \leq 0 \text{ in } \bar{D} \setminus \Gamma.$$

Case 1. $\max_{\bar{D}} \widehat{w} < 0$ done.

Case 2. $\max_{\bar{D}} \widehat{w} \geq 0$.

(a) $\max_{\bar{D}} \widehat{w}$ is attained at some $(x_*, t_*) \in \bar{D} \setminus \Gamma$

SMP (ii) $\Rightarrow \widehat{w} \equiv C$ const. for $0 \leq t \leq t_*$

I.C. $\Rightarrow \widehat{w} \leq 0$ $\Rightarrow \widehat{w} \equiv C = 0$ for $0 \leq t \leq t_*$.

But $C \geq 0$ $\Rightarrow w \equiv 0$ for $0 \leq t \leq t_*$.

in particular $u(x,0) = v(x,0)$ $\forall x \in \Omega$.

(b). $\max_{\bar{D}} \tilde{w}$ is not achieved in $\bar{D}\setminus\Gamma$, but at some $(x_0,t_0)\in\Gamma$.

(b1). $(x_0,t_0)\in S$. Hopf $\Rightarrow \frac{\partial \tilde{w}}{\partial \vec{\jmath}}(x_0,t_0)>0$.

But. $B\tilde{w} = \frac{\partial \tilde{w}}{\partial \vec{\jmath}} + \beta \tilde{w} = e^{-Mt}\cdot Bw \le 0$ on S.

At (x_0,t_0): $\quad >0 \quad \geq 0 \quad \Rightarrow\Leftarrow$.

(b2) $(x_0,t_0)\in \Omega\times\{t=0\}$. Then $\max_{\bar{D}} \tilde{w} = \tilde{w}(x_0,0)\le 0 \Rightarrow \max_{\bar{D}} \tilde{w} = 0$.

$\Rightarrow \tilde{w}\le 0$ & $w\le 0$ in $\bar{D}$.

Since $\max_{\bar{D}} \tilde{w}$ is attained only at the bottom of $\bar{D}$, i.e., $t=0$, we see $\tilde{w}$ & $w<0$ for $t>0$. $\square$.